COMP662 — Generative Artificial Intelligence

Assignment 2: Retrieval-Augmented Generation

Semester 2, 2026 | Weight: 30% of final grade | Individual Assessment

Item	Detail
Due Date	5:00 p.m., Sunday, 4 October 2026
Submission Platform	Akoraka | Learn
Assessment Type	Individual
GenAI Use	Permitted (see Section: GenAI Policy)

Late Submission Policy

Time Late	Penalty
Less than 6 hours late	5% late penalty
Between 6 and 24 hours late (up to 1 day)	12.5% late penalty
Between 24 and 48 hours late (1–2 days)	25% late penalty
Over 48 hours late (2 days or longer)	100% late penalty

Penalties are applied to the mark achieved, not the maximum available. Extensions more than 3 days require formal aegrotat application with supporting documentation before the due date.

Academic Integrity

This is an individual assessment. Each student must produce their own analysis and code. While general concepts may be discussed verbally with peers, all submitted work—including code, visualisations, and written interpretations—must be entirely your own. Submissions containing duplicated or substantially identical content will result in zero marks for all parties involved.

You must not share your code, notebook, or any written material with other students. Allowing another student to copy your work is also a breach of academic integrity.

GenAI Policy

Generative AI (GenAI) tools are permitted in this course. You may use them for brainstorming, code debugging, writing assistance, and conceptual clarification. However, you remain fully responsible for the correctness and originality of all submitted work.

You must include a GenAI Acknowledgement file stating which tools you used and for what purpose. If no GenAI tools were used, state this explicitly. Failure to declare GenAI use where it has been used may be treated as a breach of academic integrity.

Overview

You will develop an integrated Large Language Model (LLM)-based system that combines Retrieval-Augmented Generation (RAG) and prompt engineering to support preliminary land-use planning decisions in areas exposed to natural hazards. The system will focus on proposed residential, community and commercial development in Selwyn District where flooding and/or liquefaction may be relevant.

You need to develop a Natural-Hazard Land-Use Assistant that helps a landowner or junior planner undertake a preliminary assessment. Your system will retrieve evidence from a curated collection of national, regional, and district policy documents; integrate that evidence with structured site and development information; and generate a transparent preliminary assessment. It must cite the supporting document and clause, identify missing or uncertain information, and avoid presenting its output as a formal resource-consent, building-consent, legal, or engineering decision.

The purpose is not to produce the most permissive or restrictive recommendation. The emphasis is on building a technically sound RAG pipeline, optimising prompts systematically, evaluating retrieval and generation separately, and critically analysing system limitations in a high-stakes policy context.

Learning Objectives

Upon completing this assignment, you should be able to:

- Prepare and index a heterogeneous policy-document corpus while preserving document structure and source metadata.
- Design, implement, and improve a RAG pipeline using appropriate chunking, embedding, retrieval, and ranking methods.
- Develop and compare prompt-engineering strategies that produce grounded, structured, and appropriately cautious responses.
- Evaluate retrieval and generation quality using appropriate quantitative and qualitative methods.
- Critically analyse hallucination, uncertainty, source authority, professional responsibility, and other limitations of policy-facing LLM systems.

Dataset

You will be provided with a frozen teaching dataset: four policy and guidance documents totalling 21 pages and 84 marked clauses, twelve fictional site cases, and the question sets and gold labels used for evaluation. Treat the supplied versions and case facts as authoritative for this assessment. Do not scrape live websites or add external policy documents to the indexed corpus, because policy versions and online content may change during the assignment.

Every clause in every document is introduced by a marker line of the form [[CLAUSE <clause_id>]] followed by the clause heading. The clause_id in that marker is exactly the clause_id used in gold_relevance.csv, and the markers survive standard PDF text extraction. You can therefore attach a reliable clause identifier to each chunk without inferring it from heading formats.

File or Folder	Description
policy_corpus/	Four frozen national, regional and district policy and guidance documents (21 pages, 84 marked clauses) supplied for this assessment. Use only the supplied versions when building the indexed corpus.
corpus_manifest.csv	Source-level metadata for the supplied documents, including title, authority level, status, version information, and source URL.
site_cases.json	Fictional development cases containing proposal and site facts, supplied flood and liquefaction evidence, mitigation information, intentionally missing fields, and hazard-specific risk inputs where provided.
development_questions.xlsx	32 development questions, DEV001 - DEV032, with query type, case identifier, retrieval eligibility, and an expected_behavior column stating what a correct response must establish. Use these for baseline demonstrations, retrieval development and prompt development.
final_evaluation_questions.xlsx	24 reserved final evaluation questions, TEST001 - TEST024, for Task 5. Do not use these questions to tune, select, or modify the retriever or prompt.
gold_relevance.csv	Gold document-and-clause relevance mappings (195 rows over 44 distinct evidence units) used for retrieval evaluation. Relevance is defined by policy evidence units (document_id + clause_id) rather than by student-generated chunk IDs.
site_cases_schema.json	Defines the required structure and validation rules for site_cases.json.

Case Data

The cases in site_cases.json provide structured facts; the policy corpus provides the rules, objectives, policies, definitions, and guidance needed to interpret those facts. Your system must not infer site-specific hazard values from an address or from the absence of a map result.

Entries in risk_levels are hazard-specific. Each case contains separate flooding and liquefaction risk levels. Where either value is null for a hazard, your system must report that the risk level cannot be determined for that hazard, and must not substitute a value inferred from flood depth, mapped susceptibility, an address, or the absence of an overlay.

For fields such as climate_change_considered inside the flood entry:

- true = explicitly stated as considered

- false = explicitly stated as not considered

- null = not supplied / unknown

The supplied national policy statement contains a likelihood table mapping AEP and ARI to a likelihood level. If your system derives a likelihood from that table, the result is a derived estimate and must be labelled as derived, alongside a statement that the structured case facts did not supply a likelihood. It must not be presented as a factual risk level result, and it must not be used in place of a supplied risk_levels value.

Field Group	Examples
Proposal	Activity type, number of lots or buildings, expected occupancy, vulnerable occupants, design life
Site	Planning zone, existing use, supplied hazard overlays, available reports
Flood	Annual exceedance probability or return period, modelled depth, velocity, climate-change assumptions
Liquefaction	Susceptibility category, mapped evidence, availability of site-specific geotechnical assessment
Mitigation	Raised floor level, freeboard, drainage, foundation design, access or evacuation measures
risk_levels	Hazard-specific risk levels where supplied
known_uncertainties	Missing reports, conflicting evidence, unassessed neighbouring effects, incomplete climate assumptions

Required Technical Configuration

To support fair comparison and reproducibility, use the following common baseline unless an alternative is approved before submission:

- Generator: Qwen/Qwen2.5-1.5B-Instruct, run locally with Hugging Face Transformers. Use the same checkpoint and runtime configuration throughout the assignment. support/MODEL_SETUP.md gives the device and dtype settings, measured performance figures, and the generation budget for the whole assignment.
- This is the same generator used in Tutorial 9. Every student uses the 1.5B checkpoint: there is no smaller assessed alternative, and the 0.5B checkpoint offered as a fallback in Tutorial 9 is not an option for this assignment. You can run the smoke test in Section 9 of assignment2_support.ipynb to report your device and measured tokens per second.
- If your own machine cannot run the 1.5B checkpoint, or is too slow to be practical, use Google Colab or Kaggle. It runs the same checkpoint, so your results remain comparable with everyone else's. Colab's free tier does not guarantee any particular accelerator, or one at all; check what you were assigned before starting and record it in your README. support/MODEL_SETUP.md gives the Colab setup, including mounting the dataset and the model cache from Google Drive so they survive a disconnect.
- Record in your README which environment you used, local or Colab or Kaggle, together with the device and dtype, because these sit behind the timings you report.

- Baseline embedding model: BAAI/bge-small-en-v1.5.
- Vector store: FAISS.
- Baseline retrieval: dense cosine-similarity search with top-k = 5.
- Generation settings: deterministic greedy decoding (do_sample=False) with max_new_tokens = 1024. Keep all decoding settings unchanged across controlled comparisons.
- Environment: support/requirements.txt lists a version combination that course staff installed and ran end to end. You may pin different versions, but say so in your README and confirm your notebook still runs top to bottom.

You may introduce additional retrieval or reranking models in Task 3, but the prescribed generator, checkpoint, and decoding configuration must remain fixed across controlled comparisons. Do not fine-tune the LLM.

Support Materials

The following materials provide limited engineering support so that the assessment focuses on RAG design, prompt engineering, integration, and evaluation rather than on unfamiliar library APIs. They do not implement the assessed solution.

- assignment2_support.ipynb: a runnable notebook covering an environment and corpus integrity check, PDF loading, recovery of clause identifiers, BAAI/bge-small-en-v1.5, a FAISS cosine index over the supplied corpus, Recall@5 and MRR@5 helpers, a citation parser for the Task 4 measures, and a Qwen smoke test that reports your device and measured tokens per second. It does not implement the assessed solution.
- evaluation_guide.md: the required response-behaviour measures for Tasks 4 and 5, and the failure-analysis categories used in Task 5.
- MODEL_SETUP.md: the prescribed generator configuration, device and dtype guidance, measured performance figures, and the generation budget for the assignment.
- requirements.txt: a tested reference environment.

Expected Assessment Output

For each case-specific question, the final assistant should provide the following sections:

- Proposal summary and relevant supplied site facts.
- Applicable hazards and scope checks.
- Relevant national, regional, and district provisions with document-and-clause citations.
- Mitigation considerations and possible effects on neighbouring sites.
- Missing, uncertain, or conflicting information.
- A cautious preliminary recommendation and appropriate next steps.
- Limitations and a statement that the output is not a formal planning or engineering decision.

Tasks

This assignment consists of five tasks. All written analysis, experimental results, and supporting visualisations should be presented in the Jupyter notebook alongside the relevant code. Use the development questions for system development and selection. The reserved final evaluation questions are for Task 5 only and must not be used for tuning.

Task 1: Corpus Understanding and Indexing (10 marks)

Inspect the supplied policy corpus and construct a traceable document-processing pipeline. The objective is to preserve the structure and authority of policy evidence rather than treating the corpus as an undifferentiated collection of text.

You should:

- Summarise the documents by authority level, topic, status, effective date, format, and approximate length.
- Extract and clean the text while preserving headings, clause numbers, table content, and document boundaries. Inspect how tables are extracted from the supplied documents and handle any loss of row or cell structure during extraction. Justify your approach.
- Create useful chunk metadata, including document title, authority level, section or clause, page, hazard type, status, and effective date where available.
- Implement and justify a chunking strategy, including chunk size, overlap, and treatment of headings, tables, definitions, rules, and exceptions. You can also talk about any potential issues from your inspection.

Task 2: Baseline RAG System (20 marks)

Build a complete baseline RAG pipeline using the required generator, baseline embedding model, and dense top-k retrieval. This baseline provides the reference point for later improvements.

You should:

- Create embeddings for the processed chunks and store them in FAISS.
- Implement dense similarity retrieval with top-k = 5 and display the retrieved text and metadata.
- Construct a simple RAG prompt that supplies the retrieved context, the user question, and any supplied case facts required by the question to Qwen/Qwen2.5-1.5B-Instruct. Do not introduce structured prompt-engineering or other advanced techniques at this stage.
- Demonstrate the baseline on at least five development questions covering different query types. The query_type column in development_questions.xlsx takes the values direct_policy, case_specific, multi_document, missing_information, out_of_scope, source_authority and uncertainty; your five must include at least one out_of_scope and one missing_information question.

At this stage, do not optimise the prompt or retrieval pipeline. The purpose is to establish a clearly defined and reproducible baseline.

Task 3: Retrieval Enhancement and Evaluation (25 marks)

Develop and evaluate an improved retrieval pipeline for the supplied policy corpus. Your final retriever must incorporate **at least one of the two** justified improvements beyond the baseline:

- Keyword or BM25 retrieval, combined with dense retrieval.
- Similarity-score thresholding for unsupported or out-of-scope queries.

For each change, identify the weakness it is intended to address, and examine and discuss whether the results support that expectation.

Evaluate the **baseline** and **improved configurations** on development_questions.xlsx using the supplied gold relevance labels. For the hybrid and dense retrieval comparison, report Recall@5 and MRR@5 using the top five ranked results, averaged over the 29 retrieval-eligible development questions.

For thresholding, the out-of-scope development questions DEV019 and DEV020 have retrieval_eval_eligible=no and are excluded from Recall@5 and MRR@5, because a retriever that correctly declines to return results for them would otherwise be scored as a total retrieval failure for doing the right thing. Instead, measure the threshold at the retriever, using the retrieval-level abstention pair defined in evaluation_guide.md section 1.1 — correct-rejection rate and false-rejection rate — at your selected threshold and at least one other value, each as numerator / denominator = percentage (denominators 2 and 29 respectively), and show the score distributions you chose the threshold from. A threshold reported without its false-rejection rate is not proper. Both rates are properties of the retriever alone and need no generated answers. The three answer-level rates in evaluation_guide.md section 2.4 must not be used to choose a threshold here. It belongs to Tasks 4 and 5, where the assistant's own behaviour is what is being assessed. It is worth noting that the correct rejection does not mean there is no output at all. The citation output should still be available for citation coverage check in the next two tasks.

Select and freeze one final retrieval configuration for Tasks 4-5. Do not use generated-answer quality to choose between multiple retrieval configurations, and do not use final_evaluation_questions.xlsx to tune or select the retriever.

Task 4: Prompt Engineering and Grounded Response Generation (20 marks)

Design, compare, and refine prompting strategies for evidence-grounded preliminary planning assessments. Use the retriever selected in Task 3, keep the retrieved passages, generator, case facts, and generation settings fixed, so that differences can reasonably be attributed to the prompt.

Compare all configurations on the same eight-question prompt-development subset (**prompt_dev_subset=yes in development_questions.xlsx**). The subset covers seven query types, including one out-of-scope question, so a prompt cannot score well by being uniformly confident.

You must compare at least three prompt configurations:

- P1 - Basic RAG prompt: the simple prompt established in Task 2, used unchanged as the prompt-engineering baseline. Do not add a citation format to it.
- P2 - Structured grounded prompt: specify the required response structure from Expected Assessment Output, how retrieved evidence is to be used, and a citation format drawn from the forms accepted in evaluation_guide.md section 2.1 (the parenthesised form is preferred); and distinguish supplied case facts from retrieved policy evidence.
- P3 - Targeted optimised prompt: address weaknesses you actually observed in P1 and P2, using at least one named prompt-engineering technique from the list below in addition to any content instructions.

The named techniques should change only the prompt text: few-shot prompting; role prompting; structured output prompting; and chain-of-thought prompting. P3 must use at least one of these.

Content instructions such as source-authority rules, explicit scope checking, uncertainty and missing-information handling, claim-evidence separation, or an abstention rule are also expected in P3. They are instructions about what to say rather than prompt-engineering techniques. **Report your exact prompt for each question.**

For each prompt change, state the weakness it is intended to address, and examine whether the observed results support that expectation. If several techniques are introduced together, explain their individual purposes and how their effects can be separated.

For every configuration, report all of the following. Section 2 of evaluation_guide.md define each measure and how to compute it.

- Citation validity rate, with the fabricated and out-of-context counts reported separately, and the unattributed and suspicious counts reported alongside it. Each bucket is de-duplicated per answer.
- Required-evidence citation coverage, computed only over the required gold units that were actually present in the supplied context.
- The three abstention rates in evaluation_guide.md section 2.4 — scope-refusal, uncertainty-handling and false-refusal. Report each as numerator / denominator = percentage. On this eight-question subset the denominators are 1, 1 and 4, so a single answer moves a rate by 25 to 100 percentage points; a bare percentage would hide that.
- Cost: mean prompt tokens, mean generated tokens, mean seconds per answer, the device used, and the number of answers that hit max_new_tokens without terminating.

Select and freeze one prompt for Task 5. Support your selection with a difference that holds across most of the eight questions, or with a demonstrated behavioural difference shown through a specific pair of outputs. Do not use final_evaluation_questions.xlsx to refine or select the prompt.

Task 5: End-to-End Evaluation and Critical Analysis (25 marks)

Evaluate the frozen baseline and final systems on final_evaluation_questions.xlsx. This task assesses final system performance and limitations; the reserved final evaluation set must not be used to tune, select, or modify retrieval or prompt configurations.

For a fair comparison, use the same reserved questions, supplied case facts, prescribed generator, and generation settings for both systems. Define the baseline as the Task 2 dense retriever plus the P1 basic prompt (different from the Task 4 baseline), and define the final system as the retriever selected in Task 3 plus the prompt selected in Task 4.

Your evaluation of retrieval and generated responses must include:

- Recall@5 and MRR@5 for baseline and final retrieval, averaged over the 21 retrieval-eligible questions in final_evaluation_questions.xlsx.
- The response-behaviour measures in evaluation_guide.md section 2 — citation validity rate, required-evidence citation coverage, the three abstention rates, and cost — reported for both the baseline and the final system. The baseline uses the P1 prompt, which has no citation format. If it emits no checkable citations, its citation validity rate is n/a with a citation count of

zero, and its citation coverage is either zero or n/a. That is the expected result and should be reported as a finding about the baseline, not worked around.

- Detailed diagnosis of at least five important failure cases, identifying whether the primary cause arose from retrieval, context construction, prompting, generation, source conflict, insufficient information, or evaluation limitations.
- Critical discussion of hallucination, incomplete hazard data, source authority differentiation, and limits on generalising beyond the supplied teaching cases.
- Conclude by identifying where the final system is reasonably reliable, where planner, council, engineering, or other specialist review remains essential, and the single improvement that would provide the greatest practical benefit if the project were continued. Giving justification and evidence where relevant.

Do not modify the final system in response to Task 5 results. Report findings from the reserved evaluation as limitations or future improvements rather than using them for further tuning.

Submission Requirements

Create a GitHub repository containing all submission documents and code. The GitHub repository should be **private**. You must add the **lincolnmac** account as the collaborator to your GitHub repository. Submit a text file named GitHub_URL.txt through Akoraka | Learn containing only the repository URL. Name the repository **StudentID-COMP662A2**, replacing StudentID with your student ID (for example, 12345678-COMP662A2). The GitHub repository should contain the following documents:

File or Folder	Description
StudentID_Assignment2.ipynb	Notebook containing corpus analysis, preprocessing, baseline and improved RAG pipelines, prompt experiments, quantitative results, visualisations, qualitative examples, failure analysis, and critical discussion.
README.md	Pinned Python dependencies required to reproduce the submission; Concise setup and execution instructions, expected data-folder structure.
Optional extracted policy corpus	You can provide extracted policy corpus as separate files.
GenAI_Acknowledgement.txt	A statement describing the prescribed model and any additional GenAI tools used, together with the purpose of each use.

Use relative file paths throughout. You should document your generation outputs in the notebook. A marker reads those outputs rather than re-running your generations. The final notebook must run in a clean Python environment using the instructions provided.

You must not push additional commits or modify the repository after the assignment deadline unless an approved extension applies. Post-deadline changes may be treated as a late submission.

Marking Criteria

The following table outlines the marks allocated to each task and provides general descriptors for three performance bands.

Task	Marks	High (80–100%)	Adequate (50–79%)	Low (0–49%)
Task 1: Corpus Understanding and Preprocessing	10	Corpus is analysed accurately; extraction preserves policy structure; metadata and chunking are well designed and justified。	Corpus is processed successfully with generally suitable chunks and metadata; some justification and inspection are provided.	Corpus preparation is incomplete, poorly structured, or not reproducible; important metadata, clauses, or document boundaries are lost; little justification or inspection is provided.
Task 2: Baseline RAG System	20	Complete baseline integrates embedding, indexing, retrieval, context construction, and generation; settings are explicit; demonstrations cover diverse question types and reveal baseline behaviour clearly.	Functional baseline is implemented with most settings reported; demonstrations are adequate but limited in diversity, or transparency.	Baseline is incomplete or unreliable; major RAG components are missing or incorrectly connected; settings and demonstrations are insufficient.
Task 3: Retrieval Enhancement and Evaluation	25	At least one meaningful retrieval improvements are correctly implemented. Each change is motivated by a weakness and evaluated under controlled development-set conditions using Recall@5, MRR@5, and any appropriate method-specific measures. Query-level results are analysed insightfully, and the final retriever is selected by evidence and frozen before prompt experiments.	At least one reasonable retrieval changes are implemented and compared with the baseline. Required metrics are reported and the final retriever is functional, but motivation, experimental control, interpretation of individual changes, or query-level analysis is limited.	Retrieval changes are superficial, incorrectly implemented, or inadequately compared with the baseline. Required metrics or relevance labels are missing or misused, or the final configuration is selected without meaningful evidence.
Task 4: Prompt Engineering and Grounded Responses	20	At least three prompt configurations are properly compared under controlled conditions on the fixed eight-question subset, with P3 using at least one named technique. Prompt changes are motivated by mitigating weaknesses and the results are discussed. All four required response-behaviour measures are reported per configuration. The final prompt selection is justified against the small-sample limitation rather than by a bare total, and the prompt is frozen before final evaluation.	Three prompt configurations are implemented and compared under generally consistent conditions. The required measures are mostly reported, and major changes are described, but the rationale, experimental control, interpretation of individual techniques, discussion of results, or treatment of sample size is limited.	Prompt configurations differ only superficially or use no named technique, are not compared fairly, or lack a clear rationale or discussion. Required measures are missing or misapplied, evaluation relies mainly on isolated examples or unsupported impressions, and grounding, citation, uncertainty, or scope behaviour remains unreliable.
Task 5: End-to-End Evaluation and Critical Analysis	25	The frozen baseline and final systems are compared fairly on the reserved evaluation set using consistent inputs and settings. Retrieval and response quality are evaluated separately with appropriate quantitative measures. Failure cases are traced convincingly to underlying system components or evidence limitations, and the critical analysis connects observed results to professional and societal risks. The conclusion is clear and justified.	Baseline and final systems are compared using generally consistent settings and the required retrieval and response-quality measures are mostly present. Results and failures are interpreted reasonably, but evaluation control, root-cause diagnosis, or connection to observed evidence is incomplete. Critical analysis and conclusion are partial.	Final evaluation is incomplete, unfair, or contaminated by further tuning. Required measures are missing or incorrectly applied, failures lack meaningful diagnosis, or limitations are generic and disconnected from observed system behaviour. Missing or superficial performance analysis and conclusion.
Total	100			

Additional Guidance

- Gold labels in gold_relevance.csv support evaluation and development decisions, but must not be used as an inference-time lookup to supply the correct clauses.
- Expected_behavior column in the questions file is an evaluation reference and should not be passed to the generator with the question.